SAGA

SYSTÈME AUTOMATISÉ DE GESTION DES ABSENCES

Attendance is the key to success. Every missed class is a missed opportunity.

INTRODUCTION

SAGA (Système Automatisé de Gestion des Absences) is a digital solution aimed at simplifying the management of absences for students and professors.

ABOUT US

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PROJECT OVERVIEW

Problem :
The manual attendance system is ineffective, making it hard for professors to track attendance and easy for students to skip classes without consequences.

Solution :
SAGA automates attendance tracking, allowing professors to efficiently record attendance, students to justify absences digitally, and administrators to manage records seamlessly. This ensures accuracy, saves time, and reduces unjustified absences.

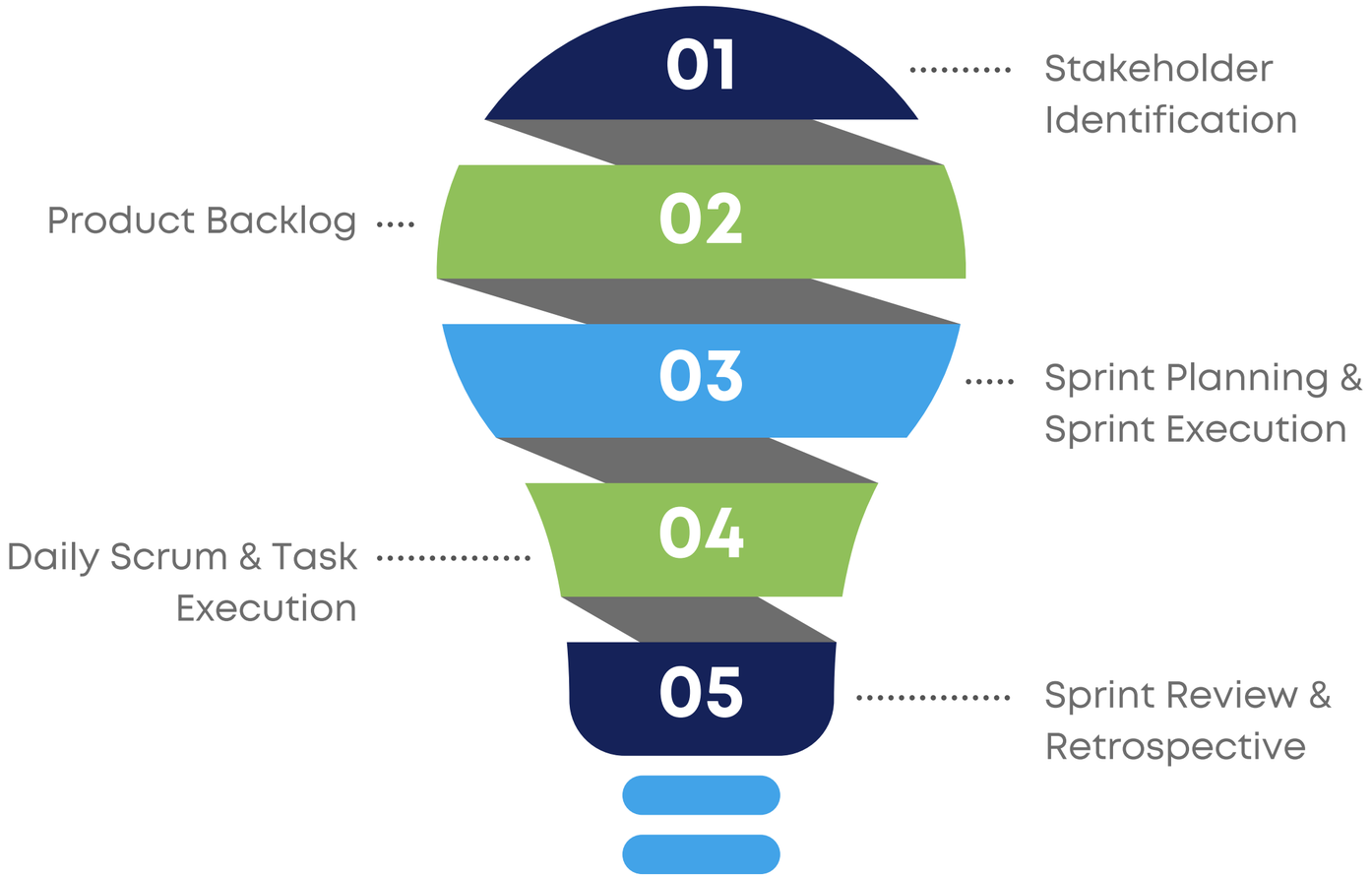
Key features :

- Automated attendence tracking
- Real-time absence monitoring
- Digital absence justification
- Centralized admin dashboard

Market :

- 286 higher education institutions in Tunisia
- Minimal competition, with no tailored local solutions

THE WORKFLOW



UI/UX

1. Color Palette & Design

The application's interface adopts a color palette dominated by dark blue and white, ensuring a professional and sleek design. This choice enhances readability and provides a modern aesthetic. Buttons and interactive elements use light blue accents to reinforce visual hierarchy.

2. Navigation & Ergonomics

A sidebar menu provides quick access to:

- Attendance dashboard
- Absence justification
- Absence management
- Timetable tracking

UI/UX

3. Graphical Elements and UI Components

The interface incorporates several essential graphical components:

- Pie charts and line graphs to visualize attendance statistics.
- Dynamic tables for managing absences and justifications.
- Intuitive forms for submitting justifications.
- Clearly identifiable action buttons ("Submit," "Approve," "Reject").

4. Accessibility and User Experience

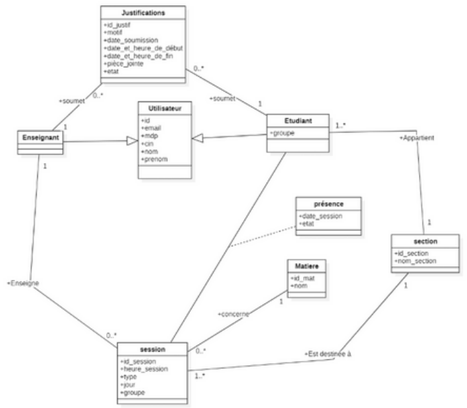
- Responsive design for all devices
- High contrast for readability
- Minimal clicks for quick actions
- Simplified forms for easy data entry

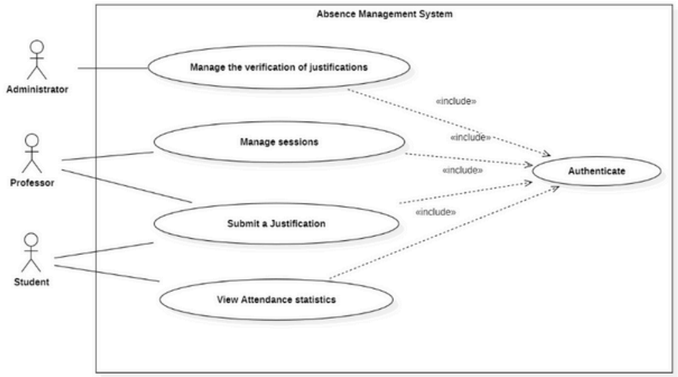
5. Figma Design Link

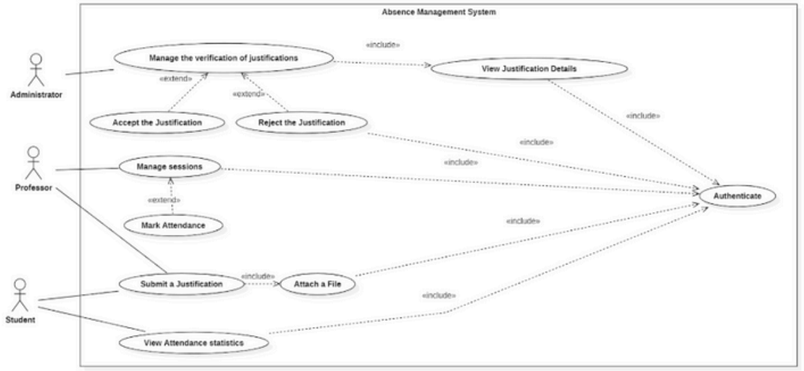
For a detailed view of the UI/UX design, you can access the Figma project here: [Figma UI/UXa Design]

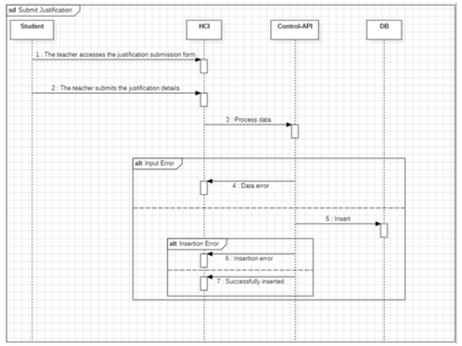
CONCEPTION

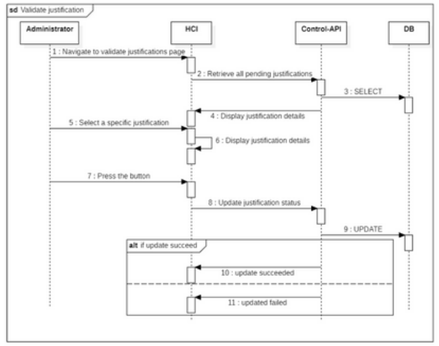
Conception defines the architecture, ergonomics, and layout of the web project. It includes planning, wireframes, diagrams, and technology selection for a solid foundation and optimal user experience.

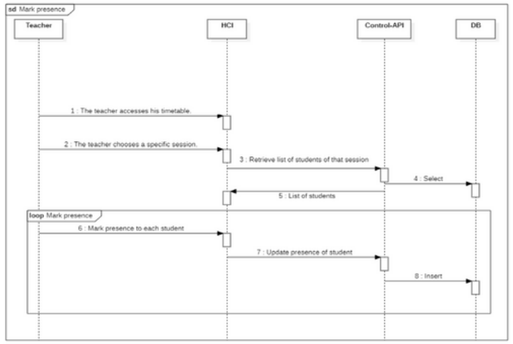


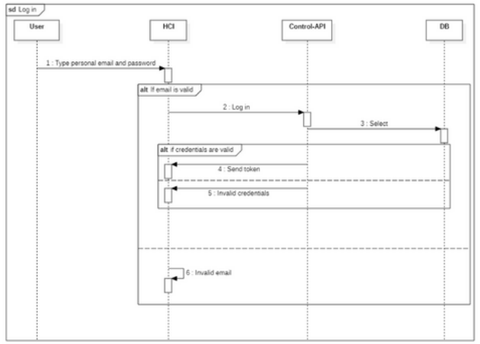












TECHNOLOGIES & LIBRAIRIES USED

To develop the SAGA system, we have chosen the following technology stack:

Technologies :

- **Frontend** : React.js
- **Backend** : Express.js
- **Database** : MySQL

Libraries :

- Recharts
- FullCalendar
- Axios

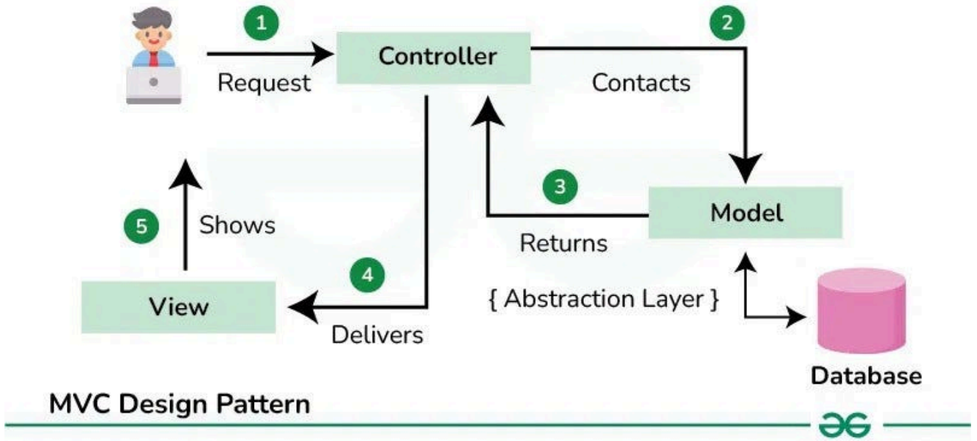
DESIGN PATTERN

1. MVC (Model-View-Controller) – Backend

2. Modular Design – Frontend

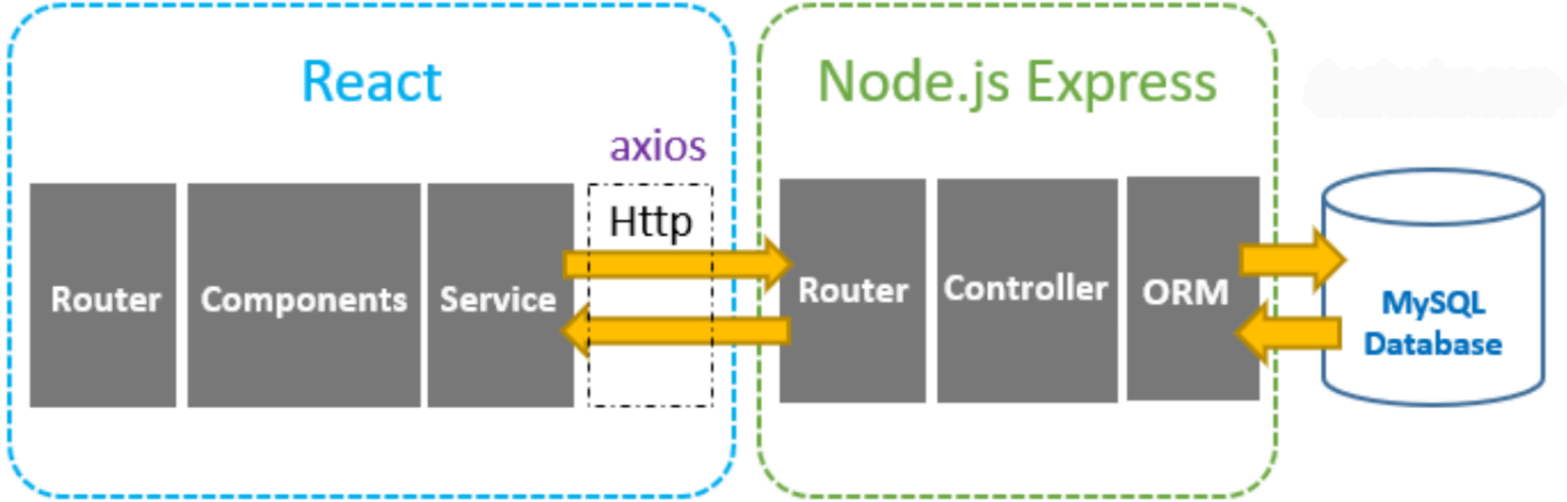
3. Service Pattern – API Calls

4. Middleware Pattern – Backend



APPLICATION ARCHITECTURE

The app follows a 3-tier model, separating presentation, logic, and data layers for better scalability.



DATABASE

MySQL was chosen for its reliability, performance, and compatibility with various web technologies. It is open-source, secure, and well-optimized for managing relational databases.

The connection between MySQL and the Express.js server was established using the mysql2 module. This module allows configuring the connection to the MySQL database by specifying the host, username, password, and database name. Once the connection is set up, Express.js can interact with MySQL to execute queries and manage data efficiently.

API

We design and implement APIs to seamlessly connect the frontend and backend of our web applications. Our APIs act as a bridge, enabling smooth communication between the user interface and the server/database. We ensure that the frontend can easily send requests, such as fetching data or submitting forms, while the backend processes these requests and responds with the necessary data. We focus on building scalable and reliable APIs that enhance the functionality and performance of our web projects.

POTENTIAL IMPROVEMENTS

- Real-time notifications
- Mobile app development
- Enhancing Student Attendance Tracking in SAGA
- Reporting
- A dashbord for the administration to insert students, professors and schedules
- AI-Based Predictive Insights